

International Cybersecurity & Digital Forensics Academy

Fellowship in Cybersecurity and Digital Forensics

Network and Service Reconnaissance with Nmap and WhatWeb

WADF104 - Cybersecurity Foundations

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Student identification

Full name	_____ WISDOM BINICHIUKU ODILI _____
Registration number	_____ 2026/FCDF/17059 _____
Linux distribution / version	_____ KALI LINUX / 2026.2 _____
VM hostname	_____ KALI _____

Lab 1: Network & Service Reconnaissance with Metasploitable 2

Nmap fundamentals, service enumeration, NSE and WhatWeb fingerprinting - fully guided for beginners

Authorisation boundary: Use only the Metasploitable 2 VM or another instructor-authorised lab target. Do not scan public systems, random IP addresses, or production infrastructure.

Lab 1 is deliberately focused on NETWORK and SERVICE reconnaissance. Web-application vulnerability testing, Burp Suite, file-upload testing, Nikto, DIRB and SQL injection are reserved for Lab 2 so that the labs do not duplicate each other.

Learning Outcomes

- Identify Kali and target IP addresses.
- Understand Nmap command structure and common scan states.
- Perform host discovery, default, version, OS, aggressive, full TCP and basic UDP scans.
- Use safe Nmap NSE scripts to enumerate services.
- Interpret ports, services, versions, banners and HTTP headers.
- Use WhatWeb at multiple aggression levels to fingerprint an authorised web service.
- Create a clear network/service inventory and evidence pack.

Part 1 - Prepare the Lab

Step 1: Find Kali IP

Objective: Know your own lab address.

```
ip addr
```

What this means: Look for the active Ethernet interface. Ignore 127.0.0.1.

What you should expect: An address such as 192.168.56.102/24.

Record in your report: Record interface and IP.

```
(kali@cyberwiz)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.101 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::2530:51c9:8e1a:ce22 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8a:35:d2 txqueuelen 1000 (Ethernet)
    RX packets 14 bytes 2103 (2.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 33 bytes 3854 (3.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

The Kali Linux Interface is **eth0** and IP is **192.168.56.101**.

Step 2: Find Metasploitable 2 IP

Objective: Identify the authorised target.

```
ifconfig
```

What this means: Run inside Metasploitable 2 and locate the active interface address.

What you should expect: An address such as 192.168.56.101.

Record in your report: Record the target IP.

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:85:97:26
          inet addr:192.168.56.102 Bcast:192.168.56.255 Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe85:9726/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500 Metric:1
          RX packets:10 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2504 (2.4 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1 Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436 Metric:1
          RX packets:93 errors:0 dropped:0 overruns:0 frame:0
          TX packets:93 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19485 (19.0 KB)  TX bytes:19485 (19.0 KB)
```

The Metasploitable 2 IP is **192.168.56.102**.

Step 3: Confirm connectivity

Objective: Verify the VMs can communicate.

```
ping -c 4 <target-ip>
```

What this means: -c 4 sends four requests then stops.

What you should expect: Replies with time= values.

Record in your report: Screenshot the successful test.

```
(kali@cyberwiz)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.078 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.064 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.047 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.046 ms

--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3050ms
rtt min/avg/max/mdev = 0.046/0.058/0.078/0.013 ms
```

```
(kali@cyberwiz)-[~]
$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=5.65 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=2.38 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=3.26 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=3.48 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3019ms
rtt min/avg/max/mdev = 2.382/3.692/5.651/1.202 ms
```

Part 2 - Understand Nmap Output

Field/State	Meaning
PORT	Port number and protocol, e.g. 22/tcp.
open	A service is accepting connections/traffic.
closed	The host is reachable but nothing is listening on that port.
filtered	Nmap cannot determine the state because filtering blocks probes/responses.
SERVICE	Nmap service identification.
VERSION	Product/version information when -sV succeeds.

Part 3 - Core Nmap Scans

Step 4: Host discovery

Objective: Discover live hosts without a normal port scan.

```
nmap -sn 192.168.56.0/24
```

What this means: -sn performs host discovery.

What you should expect: Nmap scan report entries for live lab systems.

Record in your report: Identify which result is Metasploitable 2.

```
(kali@cyberwiz)-[~]
$ nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 17:51 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0016s latency).
MAC Address: 0A:00:27:00:00:06 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00069s latency).
MAC Address: 08:00:27:9B:77:FA (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up (0.0011s latency).
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Nmap scan report for cyberwiz.local (192.168.56.101)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 6.52 seconds
```

Nmap scan report entries captured **4 live system** of the 256 IP addresses scanned of which **IP - 192.168.56.102** belongs to **Metasploitable 2**.

Step 5: Default TCP scan

Objective: Find commonly scanned open TCP ports.

```
nmap <target-ip>
```

What this means: Nmap scans its common TCP port set.

What you should expect: PORT, STATE and SERVICE columns.

Record in your report: List every open port.

```
(kali@cyberwiz)-[~]
$ nmap 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 18:06 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 1.59 seconds
```

There are **23 open ports**: 21, 22, 23, 25, 53, 80, 111, 139, 445, 512, 513, 514, 1099, 1524, 2049, 2121, 3306, 5432, 5900, 6000, 6667, 8009, 8180

Step 6: Service/version detection

Objective: Identify software behind open ports.

```
nmap -sV <target-ip>
```

What this means: -sV sends service probes and tries to identify products/versions.

What you should expect: A VERSION column with richer detail.

Record in your report: Record at least five services.

```
l- $ nmap -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 18:14 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0023s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        Netkit rshd
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8000/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.13 seconds
```

Nmap service and version detection scanned services include (FTP service, SSH service, Web server, Database): vsftpd 2.3.4, Metasploitable root shell, ProFTPD 1.3.1, OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0), MySQL 5.0.51a-3ubuntu5, PostgreSQL DB 8.3.0 - 8.3.7.

Step 7: Maximum version intensity

Objective: Understand deeper version probing.

```
nmap -sV --version-intensity 9 <target-ip>
```

What this means: Intensity 9 asks Nmap to use its full version-probe set.

What you should expect: Possibly more complete version data.

Record in your report: Compare with normal -sV.

```
l- $ nmap -sV --version-intensity 9 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 18:45 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0093s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        Netkit rshd
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8000/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.06 seconds
```

--version-intensity 9: Controls Aggressiveness of Version Detection

The intensity level determines how many probes are sent, how hard Nmap tries to identify services. Intensity level range from 0-9 with 9 being Maximum intensity. Using version intensity helps for better version identification, and use every available probe necessary to identify services as accurately as possible.

It performs a highly accurate service version detection scan against the target host, making it particularly useful for vulnerability assessment and security enumeration.

Step 8: OS detection

Objective: Estimate the target operating system.

```
sudo nmap -O <target-ip>
```

What this means: -O enables OS fingerprinting and may require elevated privileges.

What you should expect: An OS guess/fingerprint.

Record in your report: Record the OS estimate and confidence wording.

```
(kali@cyberwiz)-[~]
$ sudo nmap -O 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 18:47 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0026s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
33/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1009/tcp  open  mircoregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6067/tcp  open  irc
8009/tcp  open  ejb3
8180/tcp  open  unknown
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.93 seconds
```

Device type: general purpose

Running: Linux 2.6.X

OS CPE: cpe:/o:linux:linux_kernel:2.6

OS details: Linux 2.6.9 - 2.6.33

Step 9: Aggressive scan

Objective: Combine several discovery functions.

```
sudo nmap -A <target-ip>
```

What this means: -A combines OS detection, version detection, default NSE scripts and traceroute where applicable. It is noisier, not an instruction to attack.

What you should expect: A larger block of service/script/OS information.

Record in your report: Identify three extra findings compared with Step 5.


```

kali@cyberwiz:~$
$ nmap -A 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 22:58 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0000s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|_ STAT:
|_ FTP server status:
|_   Connected to 192.168.56.101
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsftpd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|_ 1024 60:0f:cf:e1:c0:5f:6a:7d:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_ 2048 56:56:2a:0f:21:1d:de:a7:2b:ae:d1:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ ssl-date: 2026-09-03T22:34:23+00:00; -4h24m21s from scanner time.
|_ smtp-command: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ sslv2:
|_   SSLv2 supported
|_ ciphers:
|_   SSL2_DES_192_EDE3_CBC_WITH_MD5

```

```

|_ SSL2_RC4_128_EXPORT40_WITH_MD5
|_ SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_ SSL2_RC2_128_CBC_WITH_MD5
|_ SSL2_RC4_128_WITH_MD5
|_ SSL2_DES_64_CBC_WITH_MD5
|_ ssl-cert: Subject: commonName=subuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
53/tcp    open  domain       ISC BIND 9.4.2
|_ dns-nsid:
|_ bind-version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_ http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind      2 (RPC #100000)
|_ rpcinfo:
|_   program version port/proto service
|_   100000 2 111/tcp rpcbind
|_   100000 2 111/udp rpcbind
|_   100003 2,3,4 2049/tcp nfs
|_   100003 2,3,4 2049/udp nfs
|_   100005 1,2,3 36200/udp mountd
|_   100005 1,2,3 39411/tcp mountd
|_   100021 1,3,4 39251/tcp nlockmgr
|_   100021 1,3,4 45162/udp nlockmgr
|_   100024 1 45168/udp status
|_   100024 1 49244/tcp status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath gmrregistry

```

```

1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-Subuntu5
|_ mysql-info:
|_   Protocol: 10
|_   Version: 5.0.51a-Subuntu5
|_   Thread ID: 18
|_   Capabilities flags: 43564
|_   Some Capabilities: LongColumnFlag, SupportsCompression, Support41Auth, SupportsTransactions, SwitchToSSLAfterHandshake, Speaks41ProtocolNew, ConnectWithDatabase
|_   Status: Autocommit
|_   Salt: -K#QLXW,r[3xAvT-q
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
|_ ssl-cert: Subject: commonName=subuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
|_ ssl-date: 2026-09-03T22:34:22+00:00; -4h24m21s from scanner time.
5900/tcp  open  vnc          VNC (protocol 3.3)
|_ vnc-info:
|_   Protocol version: 3.3
|_   Security types:
|_   VNC Authentication (2)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
|_ ssp-methods: Failed to get a valid response for the OPTION request
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
|_ http-title: Apache Tomcat/5.5
|_ http-favicon: Apache Tomcat
|_ http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Device type: general purpose

```

```

Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ smb-security-mode:
|_   account_used: guest
|_   authentication_level: user
|_   challenge_response: supported
|_   message_signing: disabled (dangerous, but default)
|_ clock-skew: mean: -3h24m21s, deviation: 2h00m00s, median: -4h24m21s
|_ smb-os-discovery:
|_   OS: Unix (Samba 3.0.20-Debian)
|_   Computer name: metasploitable
|_   NetBIOS computer name:
|_   Domain name: localdomain
|_   FQDN: metasploitable.localdomain
|_   System time: 2026-09-03T18:34:08-04:00
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

TRACEROUTE
HOP RTT ADDRESS
1 9.77 ms 192.168.56.102

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 41.05 seconds

```


Instead of typing multiple options, the **aggressive scan(-A)** shortcut turns on four main features: **OS Detection (-O)**: Fingerprints the target's operating system via TCP/IP packet behavior. **Version Detection (-sV)**: Identifies the exact application and version number running on open ports. **Script Scanning (-sC)**: Runs default Nmap Scripting Engine (NSE) scripts to gather extra service info. **Traceroute (--traceroute)**: Maps the network path hops used to reach the target.

Step 10: Full TCP port scan

Objective: Check ports 1-65535.

```
sudo nmap -p- <target-ip>
```

What this means: -p- means all TCP ports.

What you should expect: Potentially more ports than the default scan.

Record in your report: List additional ports, if any.

```
(kali@cyberwiz)-[~]
$ sudo nmap -p- 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 23:13 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0017s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
39251/tcp open  unknown
39411/tcp open  unknown
49244/tcp open  unknown
57758/tcp open  unknown
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 56.46 seconds
```

Unlike the **Default TCP scan**, the **full TCP port scan** produces more results; **30 open port results**.

Step 11: Full TCP + versions

Objective: Build a fuller service inventory.

```
sudo nmap -p- -sV <target-ip>
```

What this means: Combines all-port discovery and version detection.

What you should expect: Detailed inventory of discovered TCP services.

Record in your report: Use it for your final service table.

```

(kali@cyberwiz)-[~]
$ sudo nmap -p -sV 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 23:28 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0015s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd     distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
6697/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb          Ruby DRB RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
39251/tcp open  nlockmgr     1-4 (RPC #100021)
39411/tcp open  mountd       1-3 (RPC #100005)
49244/tcp open  status       1 (RPC #100004)
57758/tcp open  java-rmi     GNU Classpath grmiregistry
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 219.74 seconds

```

Step 12: Timing template

Objective: See how a local-lab scan can be accelerated.

```
sudo nmap -p- -T4 <target-ip>
```

What this means: -T4 uses a faster timing template suited to a reliable isolated lab.

What you should expect: Similar results, often faster.

Record in your report: Record scan time and compare.

```

(kali@cyberwiz)-[~]
$ sudo nmap -p- -T4 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 23:33 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0012s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
39251/tcp open  unknown
39411/tcp open  unknown
49244/tcp open  unknown
57758/tcp open  unknown
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 59.34 seconds

```

The scan was done in 59.34 seconds compared to the previous scan time of 219.74 seconds. Nmap timing templates range from: **T0 = Paranoid**, **T1 = Sneaky**, **T2 = Polite**, **T3 = Normal**, **T4 = Aggressive**, **T5 = Insane**.

Nmap timing template speeds up the scan significantly which sends packets faster, reduces waiting time between probes and completes scans quicker.

Step 13: Selected ports

Objective: Focus on known services.

```
nmap -p 21,22,23,25,80,139,445 <target-ip>
```

What this means: -p with commas scans only selected ports.

What you should expect: Only requested ports are checked.

Record in your report: Explain when focused scanning is useful.

```
(kali@cyberwiz)-[~]
$ nmap -p 21,22,23,25,80,139,445 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 23:44 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0027s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.80 seconds
```

Focused Nmap scanning is scanning only specific hosts, ports, or services of interest rather than performing a broad scan against an entire system or network. The goal is to reduce scan time, minimize network traffic, and gather detailed information about a particular target area.

It is useful for:

- Scanning Specific Ports (known ports, or services)
- Scanning a Port Range (1-1000)
- Scanning Service Version Detection on Known Ports
- Focusing on a Single Service

It can be used after initial reconnaissance to perform deeper enumeration on identified services, making the scanning process faster, more efficient, and more targeted.

Step 14: Port range

Objective: Scan a continuous range.

```
nmap -p 1-1024 <target-ip>
```

What this means: A hyphen defines an inclusive range.

What you should expect: Results for ports 1 through 1024.

Record in your report: Compare with Step 13.

```
(kali@cyberwiz)-[~]
$ nmap -p 1-1024 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 23:53 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00056s latency).
Not shown: 1012 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.32 seconds
```

Step 15: UDP scan

Objective: Discover UDP services.

```
sudo nmap -sU <target-ip>
```

What this means: UDP often takes longer and can produce open|filtered results.

What you should expect: UDP service states.

Record in your report: Record identified UDP services.

```
(kali@cyberwiz)-[~]
$ sudo nmap -sU 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-04 03:17 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0020s latency).
Not shown: 945 closed udp ports (port-unreach), 51 open|filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
111/udp   open  rpcbind
137/udp   open  netbios-ns
2049/udp  open  nfs
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1020.12 seconds
```

The UDP services discover scan takes longer to produce results.

Identified UDP services includes **domain**, **rpcbind**, **netbios-ns** and **nfs**.

Step 16: Top UDP ports + versions

Objective: Keep UDP practical for beginners.

```
sudo nmap -sU --top-ports 20 -sV <target-ip>
```

What this means: Scans a small common UDP set and identifies services.

What you should expect: A shorter UDP inventory.

Record in your report: Explain each state returned.

```

(kali@cyberwiz)-[~]
$ sudo nmap -sU --top-ports 20 -sV 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-04 03:58 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0025s latency).

PORT      STATE      SERVICE      VERSION
53/udp    open       domain       ISC BIND 9.4.2
67/udp    closed     dhcpcd
68/udp    open|filtered dhcpcd
69/udp    open|filtered tftp
123/udp   open|filtered ntp
135/udp   closed     msrpc
137/udp   open       netbios-ns   Microsoft Windows netbios-ns (workgroup: WORKGROUP)
138/udp   open|filtered netbios-dgm
139/udp   closed     netbios-ssn
161/udp   open|filtered snmp
162/udp   closed     snmptrap
445/udp   closed     microsoft-ds
500/udp   closed     isakmp
514/udp   closed     syslog
520/udp   open|filtered route
631/udp   open|filtered ipp
1434/udp  closed     ms-sql-m
1900/udp  closed     upnp
4500/udp  open|filtered nat-t-ike
49152/udp closed     unknown
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Service Info: Host: METASPLOITABLE; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 112.87 seconds

```

Instead of scanning all UDP ports (65,535 ports), Nmap scans only the 20 most commonly used ports.

The state returned are open, closed and filtered. **Open Port state** is for No response or Service-specific response, **Closed Port state** is for ICMP Port Unreachable message, and **Filtered Port state** are for Firewall blocks probes.

Part 4 - Safe Nmap NSE Enumeration

Step 17: Default scripts

Objective: Collect common service information.

```
nmap -sC -sV <target-ip>
```

What this means: -sC runs Nmap default scripts; -sV adds service versions.

What you should expect: Script output under relevant ports.

Record in your report: Record two useful script findings.

```

(kali@cyberwiz)-[~]
$ nmap -sC -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-04 05:23 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00083s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
21/tcp    open      ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to 192.168.56.101
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_  vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open      ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_  1024 60:09:cf:el:c0:5f:6a:7a:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:eb:f3 (RSA)
23/tcp    open      telnet       Linux telnetd
25/tcp    open      smtp         Postfix smtpd
|_ssl-date: 2026-09-04T01:51:16+00:00; -7h32m39s from scanner time.
|_smtp-commands: metasploit.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_sslv2:
|_SSLv2 supported

```



```

| SSLv2 supported
| cipher:
|   SSL2_RC4_128_WITH_MD5
|   SSL2_DES_192_EDE3_CBC_WITH_MD5
|   SSL2_DES_64_CBC_WITH_MD5
|   SSL2_RC2_128_CBC_WITH_MD5
|   SSL2_RC4_128_EXPORT40_WITH_MD5
|   SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2018-03-17T14:07:45
| Not valid after: 2018-04-16T14:07:45
53/tcp open domain 15C BIND 9.4.2
| dns-nsid:
|_ bind.version: 9.4.2
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-title: Metasploitable2 - Linux
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp open rpcbind 2 (RPC #100000)
|_ rpcinfo:
|   program version port/proto service
|   100003 2,3,4 2049/tcp nfs
|   100003 2,3,4 2049/udp nfs
|   100005 1,2,3 36200/udp mountd
|   100005 1,2,3 39411/tcp mountd
139/tcp open netbios-ssn Samba smbd 3.0-4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.0-20-Debian (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rshd
513/tcp open login OpenBSD or Solaris rlogind
514/tcp open shell Netkit rshd
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5

```

```

3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
|_ mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 31
|   Capabilities flags: 43564
|   Some Capabilities: Support41Auth, LongColumnFlag, SupportsTransactions, SwitchToSSLAfterHandshake, SupportsCompression, ConnectWithDatabase, Speaks41ProtocolNew
|   Status: Autocommit
|   Salt: I!hHq=zsQjRi;Xcg[[
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
|_ ssl-date: 2026-09-04T01:51:16+00:00; -7h32m40s from scanner time.
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2018-03-17T14:07:45
|_ Not valid after: 2018-04-16T14:07:45
5900/tcp open vnc VNC (protocol 3.3)
|_ vnc-info:
|   Protocol version: 3.3
|   Security types:
|   _ VNC Authentication (2)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
8009/tcp open cjp13 Apache JServ (Protocol v1.3)
|_ ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
|_ http-favicon: Apache Tomcat
|_ http-title: Apache Tomcat/5.5
|_ http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ smb-security-mode:
|   account_used: guest

```

```

|_ authentication_level: user
|_ challenge response: supported
|_ message signing: disabled (dangerous, but default)
smb-os-discovery:
|_ OS: Unix (Samba 3.0.20-Debian)
|_ Computer name: metasploitable
|_ NetBIOS computer name:
|_ Domain name: localdomain
|_ FQDN: metasploitable.localdomain
|_ System time: 2026-09-03T21:51:07-04:00
|_ clock-skew: mean: -6h32m39s, deviation: 2h00m00s, median: -7h32m39s
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.10 seconds

```

Step 18: HTTP title

Objective: Read the title of the lab web page.

```
nmap -p 80 --script http-title <target-ip>
```

What this means: The script requests the HTTP page title.

What you should expect: A title or no-title result.

Record in your report: Record exactly what Nmap shows.


```
(kali@cyberwiz)-[~]
$ nmap -p 80 --script http-title 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-04 10:57 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0036s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_ http-title: Metasploitable2 - Linux
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.87 seconds
```

Command:

`--script http-title`

This command connects to the web server and retrieves the content inside the HTML <title> tag.

*The page title can reveal: Application names, Server names, Login portals, Internal systems, Development sites
This helps during information gathering and asset identification.*

The title of the lab web page:

Metasploitable2 - Linux

Step 19: HTTP headers

Objective: Inspect response metadata.

```
nmap -p 80 --script http-headers <target-ip>
```

What this means: Shows HTTP response headers.

What you should expect: Header names/values.

Record in your report: Identify Server header if present.

```
(kali@cyberwiz)-[~]
$ nmap -p 80 --script http-headers 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-04 11:13 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0011s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_ http-headers:
|   Date: Fri, 04 Sep 2026 05:53:18 GMT
|   Server: Apache/2.2.8 (Ubuntu) DAV/2
|   X-Powered-By: PHP/5.2.4-2ubuntu5.10
|   Connection: close
|   Content-Type: text/html
|_ (Request type: HEAD)
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.82 seconds
```

Command:

`--script http-headers`

The command sends an HTTP request to the web server and retrieves the HTTP response headers.

HTTP headers contain valuable information about: Web server software, Operating system, Frameworks, Security settings, Connection behavior, Content types.

This information is useful for identifying technologies running on the server.

The Server header of the lab web page:

Apache/2.2.8 (Ubuntu) DAV/2

The web server software being used: **Apache**

Apache version number: **2.2.8**

The operating system hosting the web server: **Ubuntu**

WebDAV extension enabled: **DAV/2**

WebDAV allows: Remote file management, uploading files, Modifying web content. This can be important during security assessments because poorly configured WebDAV services sometimes expose files or allow unauthorized actions.

Step 20: SMB protocols

Objective: Inspect supported SMB protocol versions.

```
nmap -p 139,445 --script smb-protocols <target-ip>
```

What this means: Queries the SMB service safely for protocol support.

What you should expect: SMB protocol information when available.

Record in your report: Record protocols.

```
(kali@cyberwiz)-[~]
$ nmap -p 139,445 --script smb-protocols 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-04 11:23 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0017s latency).

PORT      STATE SERVICE
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)

Host script results:
| smb-protocols:
|   dialects:
|_    NT LM 0.12 (SMBv1) [dangerous, but default]

Nmap done: 1 IP address (1 host up) scanned in 0.77 seconds
```

Command:

`--script smb-protocols`

This command is used to determine which SMB (Server Message Block) protocol versions are supported by a target system.

The script checks which SMB dialects (versions) the target supports. This helps identify: Legacy SMB support, Modern SMB support, Potential security risks.

The SMB protocol is:

NT LM 0.12 (SMBv1)

NT LM 0.12 is Microsoft's historical name for **SMBv1**. SMBv1 is the first widely adopted version of the SMB protocol which is Used for file sharing, printer sharing, network browsing, windows authentication. It is flagged dangerous because SMBv1 is considered insecure.

SMBv1 has been associated with several well-known vulnerabilities. Examples include EternalBlue (MS17-010), WannaCry ransomware propagation and NotPetya attacks. Because of these risks, SMBv1 is typically disabled in modern environments.

Step 21: SSH host keys

Objective: Observe SSH server identity fingerprints.

```
nmap -p 22 --script ssh-hostkey <target-ip>
```

What this means: SSH servers publish host keys clients use to identify the server.

What you should expect: Algorithms/fingerprints.

Record in your report: Record algorithm names only.

```
(kali@cyberwiz)-[~]
$ nmap -p 22 --script ssh-hostkey 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-04 16:14 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0011s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
| ssh-hostkey:
| 1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
| 2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.55 seconds
```

Command:

`--script ssh-hotkey`

This command is used to retrieves the SSH server's public host keys. SSH servers use host keys to: prove their identity, prevent man-in-the-middle attacks, enable trusted connections.

SSH-Hostkey:

1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)

This tells us:

- Host key type = DSA
- Key length = 1024 bits
- Fingerprint shown: **60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd**

Fingerprint acts like a digital identifier for the key.

2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)

This tells us:

- Host key type = RSA
- Key length = 2048 bits
- Fingerprint shown: **56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3**

RSA is one of the most common SSH key algorithms.

Step 22: FTP banner

Objective: Compare a banner with -sV output.

```
nmap -p 21 -sV --script banner <target-ip>
```

What this means: Attempts to read service banner text.

What you should expect: Banner text if provided.

Record in your report: Compare banner and VERSION result.

```
(kali@cyberwiz)-[~]
$ nmap -p 21 -sV --script banner 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-05 01:10 -0400
Nmap scan report for 192.168.56.102
Host is up (0.035s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
|_banner: 220 (vsFTPd 2.3.4)
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org
/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.55 seconds
```

Command:

`-sV --script banner`

This command is used to identify the service running on port **21 (FTP)** and retrieve the service banner that the server presents during connection. This command combines built-in banner NSE Script and service version which gives a comprehensive result, combine both to get the parsed version information and the raw banner text.

Banner grabbing or “Service Fingerprinting” is a technique used to gather information about a network service by examining the banner it transmits when a connection is established. This banner often contains details such as the service name, version, and operating system.

Banner and VERSION result:

Banner result: 220 (vsFTPd 2.3.4)

Service Version result: 21/tcp open ftp vsftpd 2.3.4

Comparison between Banner and VERSION result

1. The primary difference lies in **raw data versus parsed identification**: a **banner result** is the unprocessed, literal text string returned by a service upon connection (e.g., 220 (vsFTPD 2.3.4)), whereas a **service version result** gives the software name and version number inferred by matching that banner against a signature database.
2. **Service version** is believed to be more reliable and have more context whereas **banner result** is considered unreliable because it can be customized leading to inaccurate or misleading information if taken alone.

Part 5 - Manual HTTP Confirmation

Step 23: Request headers with curl

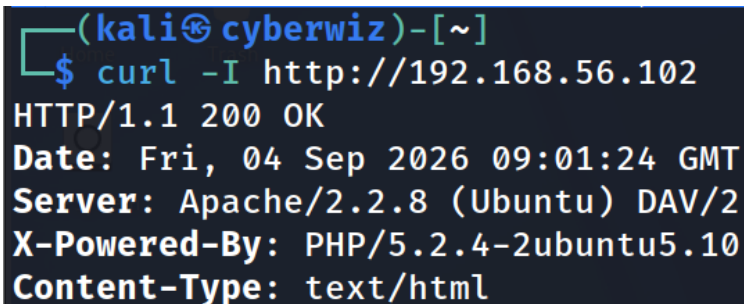
Objective: Confirm what the web service returns without a vulnerability scanner.

```
curl -I http://<target-ip>
```

What this means: -I requests headers only.

What you should expect: HTTP status and headers.

Record in your report: Record status code and Server header if present.



```
(kali@cyberwiz)-[~]  
$ curl -I http://192.168.56.102  
HTTP/1.1 200 OK  
Date: Fri, 04 Sep 2026 09:01:24 GMT  
Server: Apache/2.2.8 (Ubuntu) DAV/2  
X-Powered-By: PHP/5.2.4-2ubuntu5.10  
Content-Type: text/html
```

Status code: 200 and **Server header:** Apache /2.2.8 (Ubuntu) DAV/2

cURL (Client URL) is a command-line tool used to transfer data to and from servers using protocols such as: HTTP, HTTPS, FTP, FTPS, SCP, SFTP.

Command:

```
Curl -I
```

The -I option tells cURL to perform an **HTTP HEAD request**. Instead of retrieving the entire webpage, cURL requests only the response headers.

Part 6 - WhatWeb Fingerprinting

WhatWeb is included in Lab 1 as technology fingerprinting only. Lab 2 will use different tools for web-application testing, so this section should not be repeated there.

Step 24: Basic fingerprint

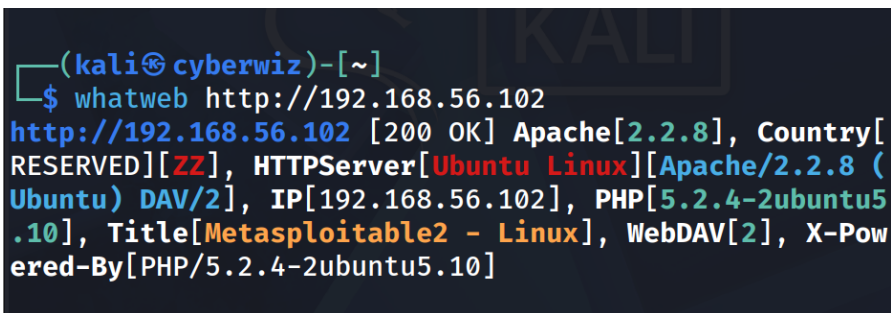
Objective: Identify exposed web technologies.

```
whatweb http://<target-ip>
```

What this means: Default fingerprinting.

What you should expect: Compact technology summary.

Record in your report: Record all recognisable technologies.



```
(kali㉿ cyberwiz)-[~]  
$ whatweb http://192.168.56.102  
http://192.168.56.102 [200 OK] Apache[2.2.8], Country[  
RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (  
Ubuntu) DAV/2], IP[192.168.56.102], PHP[5.2.4-2ubuntu5  
.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Pow  
ered-By[PHP/5.2.4-2ubuntu5.10]
```

Technologies in use:

Web Server: Apache 2.2.8

HTTP Server O/S: Ubuntu Linux

HTTP Server Module: DAV/2 (WebDav)

Scripting Language: PHP 5.2.4

Step 25: Verbose output

Objective: See plugin/fingerprint detail.

```
whatweb -v http://<target-ip>
```

What this means: -v shows verbose information.

What you should expect: More detail than the default result.

Record in your report: Compare with Step 24.


```

(kali@cyberwiz)-[~]
$ whatweb -v http://192.168.56.102
WhatWeb report for http://192.168.56.102
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.102
Country     : RESERVED, ZZ

Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
  The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

  Version      : 2.2.8 (from HTTP Server Header)
  Google Dorks : (3)
  Website      : http://httpd.apache.org/

[ HTTPServer ]
  HTTP server header string. This plugin also attempts to identify the operating system from the server header.

  OS           : Ubuntu Linux
  String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

```

```

[ PHP ]
  PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors, modules and versions and extracts the local file path and username if present.

  Version      : 5.2.4-2ubuntu5.10
  Google Dorks : (3)
  Website      : http://www.php.net/

[ WebDAV ]
  Web-based Distributed Authoring and Versioning (WebDAV) is a set of methods based on the Hypertext Transfer Protocol (HTTP) that facilitates collaboration between users in editing and managing documents and files stored on World Wide Web servers. - More Info:
  http://en.wikipedia.org/wiki/WebDAV

  Version      : 2

[ X-Powered-By ]
  X-Powered-By HTTP header

  String       : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)

```

```

HTTP Headers:
HTTP/1.1 200 OK
Date: Fri, 04 Sep 2026 09:34:13 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Connection: close
Transfer-Encoding: chunked
Content-Type: text/html

```

This option displays detailed descriptions for every detected technology, explaining exactly how the technology was identified (e.g., via HTTP headers, HTML patterns, or specific URLs).

Step 26: Aggression level 1

Objective: Understand the lightest fingerprinting mode.

```
whatweb -a 1 http://<target-ip>
```

What this means: -a controls fingerprinting aggression. Level 1 keeps requests minimal.

What you should expect: Basic fingerprints.

Record in your report: Record findings.

```
(kali@cyberwiz)-[~]
$ whatweb -a 1 http://192.168.56.102
http://192.168.56.102 [200 OK] Apache[2.2.8], Country[RESERVED], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.102], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
```

Step 27: Aggression level 3

Objective: Use additional requests in the isolated lab.

```
whatweb -a 3 -v http://<target-ip>
```

What this means: Level 3 can make more requests to improve identification.

What you should expect: Potentially richer fingerprints.

Record in your report: Compare level 1 and 3.

```
(kali@cyberwiz)-[~]
$ whatweb -a 3 -v 192.168.56.102
ERROR Opening: https://192.168.56.102 - Connection refused - connect(2) for "192.168.56.102" port 443
WhatWeb report for http://192.168.56.102
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.102
Country     : RESERVED, ZZ
Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
    The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

    Version      : 2.2.8 (from HTTP Server Header)
    Google Dorks : (3)
    Website      : http://httpd.apache.org/

[ HTTPServer ]
    HTTP server header string. This plugin also attempts to identify the operating system from the server header.

    OS           : Ubuntu Linux
    String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
    PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors, modules and versions and extracts the local file path and username if present.

    Version      : 5.2.4-2ubuntu5.10
    Version      : 5
    Google Dorks : (3)
    Website      : http://www.php.net/

[ WebDAV ]
    Web-based Distributed Authoring and Versioning (WebDAV) is a set of methods based on the Hypertext Transfer Protocol (HTTP) that facilitates collaboration between users in editing and managing documents and files stored on World Wide Web servers. - More Info:
    http://en.wikipedia.org/wiki/WebDAV
```

```

Version      : 2
[ X-Powered-By ]
X-Powered-By HTTP header
HTTP String  : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)
HTTP Headers:
HTTP/1.1 200 OK
Date: Sat, 05 Sep 2026 19:47:13 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html

```

Compared to aggressive level 1 this option provides more information.

Step 28: Aggression level 4

Objective: Observe the heaviest WhatWeb mode used in this lab.

```
whatweb -a 4 -v http://<target-ip>
```

What this means: Level 4 can generate substantially more requests; use only against the authorised lab target.

What you should expect: Potentially more/slower results.

Record in your report: State whether it added new information.

```

(kali@cyberwiz)-[~]
$ whatweb -a 4 -v http://192.168.56.102
WhatWeb report for http://192.168.56.102
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.102
Country     : RESERVED, ZZ
Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], Matomo, PHP[5.5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

```

```

Version      : 2.2.8 (from HTTP Server Header)
Google Dorks: (3)
Website      : http://httpd.apache.org/

[ HTTPServer ]
HTTP server header string. This plugin also attempts to
identify the operating system from the server header.

These are both attached to the NAT network mode, which assigns each
guest the default shared IP ( 10.0.2.15 ) via your host.

OS           : Ubuntu Linux
String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ Matomo ]
Matomo is the leading open alternative to Google Analytics
that gives you full control over your data. Matomo lets you
easily collect data from websites, apps & the IoT and
visualise this data and extract insights. Privacy is
built-in. Matomo was formerly known as Piwik, and is
developed in PHP.

Aggressive function available (check plugin file or details).
Google Dorks: (1)

```

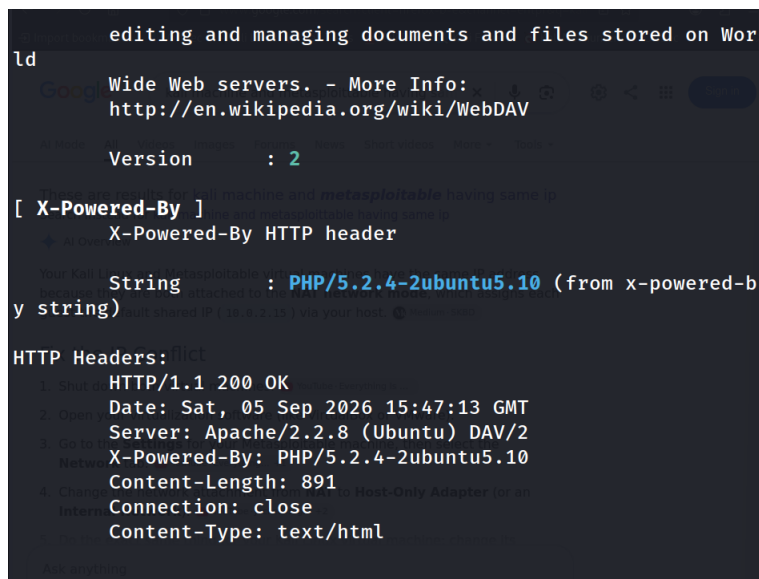
```

[ PHP ]
PHP is a widely-used general-purpose scripting language
that is especially suited for Web development and can
be embedded into HTML. This plugin identifies PHP errors,
modules and versions and extracts the local file path
and username if present.

Version      : 5.2.4-2ubuntu5.10
Version      : 5
Google Dorks: (3)
Website      : http://www.php.net/

[ WebDAV ]
Web-based Distributed Authoring and Versioning (WebDAV)
is a set of methods based on the Hypertext Transfer Protocol
(HTTP) that facilitates collaboration between users in

```



New information added: Matomo

Step 29: Follow redirects

Objective: Fingerprint the destination of redirects.

```
whatweb --follow-redirect=always http://<target-ip>
```

What this means: Follows HTTP redirect responses.

What you should expect: Fingerprinting after redirect, if any.

Record in your report: Record whether a redirect occurred.

```
(kali@cyberwiz)-[~]
$ whatweb --follow-redirect=always http://192.168.56.102
http://192.168.56.102 [200 OK] Apache[2.2.8], Country[RESERVED]
[ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2],
IP[192.168.56.102], PHP[5.2.4-2ubuntu5.10], Title[Metasploitab
le2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
```

A redirect did not occur.

A redirect would normally return one of these HTTP status codes: 301 Moved Permanently, 302 Found, 303 See Other, 307 Temporary Redirect, 308 Permanent Redirect

Since the response is **200 OK**, the server returned the requested page directly rather than instructing the client to go elsewhere.

Step 30: Save results

Objective: Preserve evidence.

```
whatweb -v http://<target-ip> > whatweb-results.txt
cat whatweb-results.txt
```

What this means: > writes output to a file; cat displays it again.

What you should expect: A saved text report.

Record in your report: Screenshot the command and file output.

```
(kali@cyberwiz)-[~]  
$ whatweb -v http://192.168.56.102 > whatweb-results.txt
```

File redirection to send result of the scan to whatweb-results.txt

Verification of whatweb-results.txt

```
(kali@cyberwiz)-[~]  
$ cat whatweb-results.txt  
WhatWeb report for http://192.168.56.102 : 192.168.56.102 / WebDAV  
Status : 200 OK  
Title : Metasploitable2 - Linux  
IP : 192.168.56.102  
Country : RESERVED, ZZ  
  
Summary : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]  
  
Detected Plugins:  
[ Apache ]  
The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.  
Version : 2.2.8 (from HTTP Server Header)  
Google Dorks: (3)  
Website : http://httpd.apache.org/  
  
[ HTTPServer ]  
HTTP server header string. This plugin also attempts to identify the operating system from the server header.
```

```
OS : Ubuntu Linux  
String : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)  
  
[ PHP ]  
PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors, modules and versions and extracts the local file path and username if present.  
Version : 5.2.4-2ubuntu5.10  
Google Dorks: (3)  
Website : http://www.php.net/  
  
[ WebDAV ]  
Web-based Distributed Authoring and Versioning (WebDAV) is a set of methods based on the Hypertext Transfer Protocol (HTTP) that facilitates collaboration between users in editing and managing documents and files stored on World Wide Web servers. - More Info:  
http://en.wikipedia.org/wiki/WebDAV  
  
Version : 2
```

```
[ X-Powered-By ]  
X-Powered-By HTTP header  
String : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)  
  
HTTP Headers:  
HTTP/1.1 200 OK  
Date: Sat, 05 Sep 2026 16:00:28 GMT  
Server: Apache/2.2.8 (Ubuntu) DAV/2  
X-Powered-By: PHP/5.2.4-2ubuntu5.10  
Content-Length: 891  
Connection: close  
Content-Type: text/html
```


Part 7 - Evidence & Reporting

- Screenshot of target identification and ping.

```
(kali@cyberwiz)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.101 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::2530:51c9:8e1a:ce22 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8a:35:d2 txqueuelen 1000 (Ethernet)
    RX packets 14 bytes 2103 (2.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 33 bytes 3854 (3.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

The Kali Linux Interface is **eth0** and IP is **192.168.56.101**.

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:85:97:26
          inet addr:192.168.56.102 Bcast:192.168.56.255 Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe85:9726/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:10 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2504 (2.4 KB) TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1 Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:93 errors:0 dropped:0 overruns:0 frame:0
          TX packets:93 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19485 (19.0 KB) TX bytes:19485 (19.0 KB)
```

The Metasploitable 2 IP is **192.168.56.102**.

```
(kali@cyberwiz)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.078 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.064 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.047 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.046 ms

--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3050ms
rtt min/avg/max/mdev = 0.046/0.058/0.078/0.013 ms
```

```
(kali@cyberwiz)-[~]
$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=5.65 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=2.38 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=3.26 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=3.48 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3019ms
rtt min/avg/max/mdev = 2.382/3.692/5.651/1.202 ms
```

- Host-discovery result.

```
(kali@cyberwiz)-[~]
$ nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 17:51 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0016s latency).
MAC Address: 0A:00:27:00:00:06 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00069s latency).
MAC Address: 08:00:27:9B:77:FA (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up (0.0011s latency).
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Nmap scan report for cyberwiz.local (192.168.56.101)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 6.52 seconds
```

Nmap scan report entries captured **4 live system** of the 256 IP addresses scanned of which **IP - 192.168.56.102** belongs to **Metasploitable 2**.

- Default Nmap result.

```
(kali@cyberwiz)-[~]
$ nmap 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 18:06 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 1.59 seconds
```

- -sV result.

```
$ nmap -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 18:14 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0021s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet   Linux telnetd
25/tcp    open  smtp     Postfix smtpd
53/tcp    open  domain   ISC BIND 9.4.2
80/tcp    open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind  2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec     netkit-rsh rshexec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  java-rmi  GNU Classpath gmiiregistry
1524/tcp  open  bindshell Metasploitable root shell
2049/tcp  open  nfs      2-4 (RPC #100000)
2121/tcp  open  ftp      ProFTPD 1.3.1
3306/tcp  open  mysql    MySQL 5.0.51a-Jubuntu5
5432/tcp  open  postgresql PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc      VNC (protocol 3.3)
6000/tcp  open  X11      (access denied)
6667/tcp  open  irc      UnrealIRCd
8009/tcp  open  ajp13    Apache Jserv (Protocol v1.3)
8180/tcp  open  http     Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.13 seconds
```

- -O and -A result.

```
(kali@kali:~)$ sudo nmap -O 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 10:47 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0026s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1399/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.93 seconds
```

- Full TCP scan evidence.

```
(kali@kali:~)$ sudo nmap -p- 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 23:13 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0037s latency).
Not shown: 65585 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1399/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8080/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgrvr
39251/tcp open  unknown
39411/tcp open  unknown
49244/tcp open  unknown
57758/tcp open  unknown
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 56.46 seconds
```

- UDP scan evidence.

```
(kali@kali:~)$ sudo nmap -sU 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-04 03:17 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0020s latency).
Not shown: 945 closed udp ports (port-unreach), 51 open/filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
111/udp   open  rpcbind
137/udp   open  netbios-ns
2049/udp  open  nfs
MAC Address: 08:00:27:85:97:26 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 1020.12 seconds
```

- At least two NSE enumeration results.

```
(kali@cyberwiz)-[~]
$ nmap -sC -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-04 05:23 -0400
Nmap scan report for 192.168.56.102
Host is up (5.000ms latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ftp           vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_  STAT
|_  FTP server status:
|_    Connected to 192.168.56.102
|_    Logged in as ftp
|_    TYPE: ASCII
|_    No session bandwidth limit
|_    Session timeout in seconds is 300
|_    Control connection is plain text
|_    Data connections will be plain text
|_    vsFTPd 2.3.4 - secure, fast, stable
|_End of status
23/tcp    open  ssh           OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_  1024 68:0f:cf:et:c8:5f:8a:7a:0b:9b:2a:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:9f:23:16:de:a7:20:ee:81:b1:3a:1d:eb:f3 (RSA)
25/tcp    open  telnet       Linux telnetd
28/tcp    open  smtp         Postfix smtpd
|_ssl-date: 2026-09-04T05:23:16-0400-00: -7h32m39s from scanner time.
|_smtp-command: METASPLOITABLE.localdomain, PDP1LINK, SIZE 10240000, VERIFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, BSMTPIME, DSN
|_sslv2:
|_  SSLv2 supported
|_  cipher:
|_    SSL2_RSA_WITH_MD5
|_    SSL2_DSS_WITH_MD5
|_    SSL2_DSS_EXPORT_WITH_MD5
|_    SSL2_RSA_EXPORT_WITH_MD5
|_    SSL2_RSA_EXPORT_WITH_MD5
|_    SSL2_DSS_EXPORT_WITH_MD5
|_  ssl-cert: Subject: commonName=ubuntu-base.localdomain/organizationName=OCDSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_  Not valid before: 2018-03-17T14:07:45
|_  Not valid after: 2018-04-10T14:07:45
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-uid:
|_  bind-version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_  program version port/proto service
|_  100000 2.2.4 2046/tcp nfs
|_  100003 2.2.4 2046/tcp nfs
|_  100005 1.2.3 10204/tcp mountd
|_  100005 1.2.3 39461/tcp mountd
139/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
1445/tcp  open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login        OpenSSH or telnetd login
514/tcp   open  shell        netkit-rsh rshd
3899/tcp  open  java-rmi     GNU Classpath gsjregistry
515/tcp   open  bindshell    Metasploit32012 root shell
2049/tcp  open  nfs          2-4 (RPC #100000)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.6.51a-Debian5
1886/tcp  open  mysql        MySQL 5.6.51a-Debian5
|_mysql-info:
|_  Protocol: 10
|_  Version: 5.6.51a-Debian5
|_  Thread ID: 31
|_  Capabilities Flags: 42006
|_  Some Capabilities: SupportsAuth, LongColumnFlag, SupportsTransactions, SwitchToSSLAfterHandshake, SupportsCompression, ConnectWithDatabase, SpeaksProtocolMw
|_  Status: Autocommit
|_  Salt: [0b04-pyl][M][M]
5432/tcp  open  postgresql   PostgreSQL 9.0.2.0 - 9.3.7
|_ssl-date: 2026-09-04T05:23:16-0400-00: -7h32m39s from scanner time.
|_ssl-cert: Subject: commonName=ubuntu-base.localdomain/organizationName=OCDSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_  Not valid before: 2018-03-17T14:07:45
|_  Not valid after: 2018-04-10T14:07:45
5900/tcp  open  vnc           VNC (protocol 3.3)
|_vnc-info:
|_  Protocol version: 3.3
|_  Security Types:
|_  VNC Authentication (2)
5901/tcp  open  x11          (access denied)
5907/tcp  open  irc          Unreal3DM2
8089/tcp  open  ajp13        Apache jkerv (protocol v1.3)
2.30-method: failed to get a valid response for the OPTIONS request
8086/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
|_http-features: Apache Tomcat
|_http-title: Apache Tomcat/2.3
|_http-server-header: Apache-Coyote/2.3
MAC Address: 08:00:27:85:97:28 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: METASPLOITABLE.localdomain, irc.METASPLOITABLE.LAM; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
Host script results:
|_smb-time: Protocol negotiation failed (SMB2)
|_smb-security-mode:
|_  account used: guest
|_  authentication_level: user
|_  challenge response: supported
|_  message signing: disabled (dangerous, but default)
|_smb-os-discovery:
|_  OS: Unix (Samba 3.0.20-Debian)
|_  Computer name: metasploitable
|_  NetBIOS computer name:
|_  Domain name: localdomain
|_  FQDN: metasploitable.localdomain
|_  System time: 2026-09-03T21:51:07-04:00
|_clock-skew: mean: -6h32m39s, deviation: 2h00m00s, median: -7h32m39s
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.18 seconds
```

- WhatWeb basic and aggression-level comparison.

Basic level

```
(kali@cyberwiz)-[~]
$ whatweb http://192.168.56.102
http://192.168.56.102 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.102], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
```


Aggression-level

```
(kali@cyberwiz)-[~]
$ whatweb -a 4 -v http://192.168.56.102
WhatWeb report for http://192.168.56.102
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.102
Country     : RESERVED, ZZ

Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2
.8 (Ubuntu) DAV/2], Matomo, PHP[5.5.24-2ubuntu5.10], WebDAV[2
], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
    The Apache HTTP Server Project is an effort to develop
and
    maintain an open-source HTTP server for modern operati
ng
    systems including UNIX and Windows NT. The goal of thi
s
    project is to provide a secure, efficient and extensib
le
    server that provides HTTP services in sync with the cu
rrent
    HTTP standards.
```

```
Version      : 2.2.8 (from HTTP Server Header)
Google Dorks : (3)
Website      : http://httpd.apache.org/

[ HTTPServer ]
    HTTP server header string. This plugin also attempts t
o
    identify the operating system from the server header.

    OS        : Ubuntu Linux
    String     : Apache/2.2.8 (Ubuntu) DAV/2 (from serve
r string)

[ Matomo ]
    Matomo is the leading open alternative to Google Analy
tics
    that gives you full control over your data. Matomo let
s you
    easily collect data from websites, apps & the IoT and
visualise this data and extract insights. Privacy is
built-in. Matomo was formerly known as Piwik, and is
developed in PHP.

    Aggressive function available (check plugin file or de
tails).
    Google Dorks: (1)
```

```
[ PHP ]
PHP is a widely-used general-purpose scripting language
that is especially suited for Web development and can
be embedded into HTML. This plugin identifies PHP errors,
modules and versions and extracts the local file path
and username if present.

Version      : 5.2.4-2ubuntu5.10
Version      : 5
Google Dorks: (3)
Website      : http://www.php.net/
```

```
[ WebDAV ]
Web-based Distributed Authoring and Versioning (WebDAV)
) is a set of methods based on the Hypertext Transfer Protocol
col (HTTP) that facilitates collaboration between users in
```

```
editing and managing documents and files stored on World
Wide Web servers. - More Info:
http://en.wikipedia.org/wiki/WebDAV
```

```
Version      : 2
```

```
[ X-Powered-By ]
X-Powered-By HTTP header
```

```
String      : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)
```

```
HTTP Headers:
HTTP/1.1 200 OK
Date: Sat, 05 Sep 2026 15:47:13 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html
```


- Final service inventory.

Port	Protocol	State	Service	Version/Evidence	Command
Host-discovery result.		Up	Metasploitable 2	Screenshot attached above.	nmap -sn 102.168.56.0/24
21, 22, 23, 25, 53, 80, 111, 139, 445, 512, 513, 514, 1099, 1524, 2049, 2121, 3306, 5432, 5900, 6000, 6667, 8009, 8180	tcp	open	ftp, ssh, telnet, Sntp, domain, http, rpcbind, netbios-ssn, microsoft-ds, exec, login, shell, rmiregistry, ingreslock, nfs, ccproxy-ftp, mysql, postgresql, vnc, X11, irc, ajp13, unknown(respectively)	Screenshot attached above.	nmap 192.168.56.102
21, 22, 23, 25, 53, 80, 111, 139, 445, 512, 513, 514, 1099, 1524, 2049, 2121, 3306, 5432, 5900, 6000, 6667, 8009, 8180	tcp	open	ftp, ssh, telnet, Sntp, domain, http, rpcbind, netbios-ssn, microsoft-ds, exec, login, shell, rmiregistry, ingreslock, nfs, ccproxy-ftp, mysql, postgresql, vnc, X11, irc, ajp13, unknown(respectively)	vsftpd 2.3.4, OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0), Linux telnetd, Postfix smtpd ISC BIND 9.4.2, Apache httpd 2.2.8 ((Ubuntu) DAV/2), 2 (RPC #100000), Samba smbd 3.X - 4.X (workgroup: WORKGROUP), Samba smbd 3.X - 4.X (workgroup: WORKGROUP), netkit-rsh rexecd, Nil, Netkit rshd, GNU Classpath grmiregistry, Metasploitable root shell, 2-4 (RPC #100003), ProFTPD 1.3.1, MySQL 5.0.51a-3ubuntu5, PostgreSQL DB 8.3.0 - 8.3.7, VNC (protocol 3.3), (access denied), UnrealIRCd, Apache Jserv (Protocol v1.3), Apache, Tomcat/Coyote JSP engine 1.1	Nmap -sV 192.168.56.102

21, 22, 23, 25, 53, 80, 111, 139, 445, 512, 513, 514, 1099, 1524, 2049, 2121, 3306, 5432, 5900, 6000, 6667, 8009, 8180	tcp	open	ftp, ssh, telnet, Sntp, domain, http, rpcbind, netbios-ssn, microsoft-ds, exec, login, shell, rmiregistry, ingreslock, nfs, ccproxy-ftp, mysql, postgresql, vnc, X11, irc, ajp13, unknown(respectively)	Device type: general purpose Running: Linux 2.6.X OS CPE: cpe:/o:linux:linux_kernel:2.6 OS details: Linux 2.6.9 - 2.6.33	sudo nmap -O 192.168.56.102
21, 22, 23, 25, 53, 80, 111, 139, 445, 512, 513, 514, 1099, 1524, 2049, 2121, 3306, 5432, 5900, 6000, 6667, 8009, 8180	tcp	open	ftp, ssh, telnet, Sntp, domain, http, rpcbind, netbios-ssn, microsoft-ds, exec, login, shell, rmiregistry, ingreslock, nfs, ccproxy-ftp, mysql, postgresql, vnc, X11, irc, ajp13, unknown(respectively)	Screenshot attached above.	nmap -A 192.168.56.102
21, 22, 23, 25, 53, 80, 111, 139, 445, 512, 513, 514, 1099, 1524, 2049, 2121, 3306, 3632, 5432, 5900, 6000, 6667, 6697, 8009, 8180, 8787, 44779, 46058, 55274, 57138	tcp	open	ftp, ssh, telnet, smtp, domain, http, rpcbind, netbios-ssn, microsoft-ds, exec, login, shell, rmiregistry, ingreslock, nfs, ccproxy-ftp, mysql, distccd, postgresql, vnc, X11, irc, ircs-u, ajp13, unknown, msgsrvr, unknown, unknown, unknown, unknown	Screenshot attached above.	sudo nmap -p- 192.168.56.102
53, 111, 137, 2049	Udp	open	domain, rpcbind, netbios-ns and nfs.	Screenshot attached above.	sudo nmap -sU 192.168.56.102

21, 22, 23, 25, 53, 80, 111, 139, 445, 512, 513, 514, 1099, 1524, 2049, 2121, 3306, 5432, 5900, 6000, 6667, 8009, 8180	tcp	open	ftp, ssh, telnet, Sntp, domain, http, rpcbind, netbios-ssn, microsoft-ds, exec, login, shell, rmieregistry, ingreslock, nfs, ccproxy-ftp, mysql, postgresql, vnc, X11, irc, ajp13, unknown(respectively)	Screenshot attached above.	nmap -sC -sV 192.168.56.102
				Screenshot attached above.	whatweb http://192.168.56.102
443				Screenshot attached above.	whatweb -a 3 -v http://192.168.56.102
				Screenshot attached above.	whatweb -a 4 -v http://192.168.56.102

Part 8 - Student Questions

1. What is reconnaissance?

Reconnaissance involves gathering information about a target before launching an attack. Performing reconnaissance enables threat actors to identify weak points in an defense system, tailor attack strategies, and increase the likelihood of a successful breach.

There are two types of reconnaissance namely active and passive reconnaissance.

Gathering information about a target without actively interacting with its systems is known as **passive reconnaissance**. Open-source intelligence (OSINT) collection, which uses publicly accessible data from websites, social media, job listings, and other online sources to provide digital footprint of a target. A target's digital footprint can also be found by using tools like Shodan and Censys, which search the internet for open ports, services, and advertising.

Active reconnaissance involves actively interacting with the target's systems and networks directly. Common techniques include:

- Port Scanning – Target networks are scanned, open ports are found, and services operating on those ports are found in this stage using tools like Nmap. Attackers can better comprehend the attack surface and possible entrance locations with the use of this information.
- Vulnerability Scanning – To find flaws in the target's software and configurations, vulnerability scanners like Burp Suite, Nmap, Nessus and OpenVAS are used. This stage is essential for identifying exploitable vulnerabilities.

- Enumeration – Attackers often use tools like Shodan and Censys enumeration tools to extract valuable data, such as user accounts, network shares, and system configurations, from target systems.

2. What is the difference between host discovery and port scanning?

In order to determine which systems on a network are active (live) and responsive, **host discovery**, whereas **port scanning** queries particular ports on those hosts to find open services and applications.

3. Explain open, closed and filtered.

Open Port state is for No response or Service-specific response, **Closed Port state** is for ICMP Port Unreachable message, and **Filtered Port state** are for Firewall blocks probes.

4. What does -sV do?

Version Detection (-sV): Identifies the exact application and version number running on open ports.

5. What does -O do?

OS Detection (-O): Fingerprints the target's operating system via TCP/IP packet behavior.

6. Explain what -A combines and why it is noisier.

The **aggressive scan(-A)** flag turns on four main features: **OS Detection (-O):** Fingerprints the target's operating system via TCP/IP packet behavior. **Version Detection (-sV):** Identifies the exact application and version number running on open ports. **Script Scanning (-sC):** Runs default Nmap Scripting Engine (NSE) scripts to gather extra service info. **Traceroute (--traceroute):** Maps the network path hops used to reach the target.

It is noisier because it combines multiple intensive scanning techniques into a single command, generating a lot more network traffic than a standard scan.

7. What does -p- mean?

The **-p-** flag in **Nmap** option instructs the scanner to **scan all 65,535 TCP ports** on a target host, rather than the default 1,000 most common ports.

8. Why can UDP scans take longer?

Unlike TCP, UDP does not use the three-handshake process (SYN, SYN-ACK, and ACK) to establish a connection instead it uses a connectionless protocol that lacks immediate feedback mechanisms. Which ends up making UDP scanning slower, harder and often less reliable.

9. What does -sC do?

Script Scanning (-sC): Runs default Nmap Scripting Engine (NSE) scripts to gather extra service info like detecting common vulnerabilities, retrieve service versions, and check for default configurations on target hosts.

10. What information did the HTTP title/headers scripts reveal?

The page title can reveal: Application names, Server names, Login portals, Internal systems, Development sites This helps during information gathering and asset identification.

HTTP headers contain valuable information about: Web server software, Operating system, Frameworks, Security settings, Connection behavior, Content types.

11. What did SMB/SSH/FTP enumeration add?

The SMB enumeration displays the dialects (versions) which the target supports. This helps identify legacy SMB support, Modern SMB support, Potential security risks.

12. What is the difference between Nmap service detection and WhatWeb fingerprinting?

Nmap Service Detection operates at the transport and application protocol levels to identify **services and their versions** on open TCP/UDP ports.

WhatWeb fingerprinting, in contrast, is specialized for identifying **web applications and CMS platforms** by analyzing HTTP responses.

13. How did WhatWeb levels 1, 3 and 4 differ?

Level	Name	Requests	Accuracy	Stealth
1	Stealthy	Usually one HTTP request per target. Relies mainly on passive fingerprinting. Fastest scan.	Basic	High
3	Aggressive	Multiple requests Performs deeper fingerprinting.	Better	Medium
4	Heavy	Generates significantly more requests and have the highest likelihood of discovering versions and hidden components. It is Noisy and easily visible in web logs.	Best	Low

14. Why must aggressive scanning/fingerprinting remain inside the authorised lab?

Because performing aggressive scanning/fingerprinting generates significant traffic that could trigger security monitoring, impact systems resources, or violate authorization rules when directed at real-world targets. So, Labs are designed for the purpose of learning the techniques.

Conclusion

After completing Lab 1, I gained knowledge and practical application for:

- Identifying Kali and target IP addresses.
- Nmap command structure and common scan states.
- Performing host discovery, default, version, OS, aggressive, full TCP and basic UDP scans.
- Using safe Nmap NSE scripts to enumerate services.
- Interpreting ports, services, versions, banners and HTTP headers.
- Using WhatWeb at multiple aggression levels to fingerprint an authorised web service.
- Creating a clear network/service inventory and evidence pack.